COMPOSITION OF FIRST-ORDER AUTOMORPHISMS ADDS THE CORRESPONDING DERIVATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Proof complete in-span; the first-order expansion of c is restated inside the proof, so the dependence on the uncarded in-chapter lemma first-order-structure-c is only for the displayed expansion and is declared as external.

Allowed standard prerequisites (editorial).

- Derivations from automorphisms of the trivial groupoid. More on Morphisms of Spaces, lemma action-by-derivations (cross-chapter, named). Identifies sections with derivations D = theta circ d_{R0/U0}.
- Local rings A, B, C and the square-zero ideal K. the local structure section of this chapter (same chapter, structural anchor). Stalk-level coordinates in which the addition is computed.
- Identity action on the conormal sheaf. lemma-idenity-on-conormal (same chapter, carded L26635). Kills the conjugation term (delta_2 circ t) acting on K in the final addition step.

Source reference. Original equation 08NB: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 472–472. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0CKE: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 579–579. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CK4: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 379–379. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CKD: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 504–504. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 03DK: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 329–329. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04D0: focused-seeds/geometry/S9-smg-composition-is-addition/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The bijection (0CKE) is an isomorphism of groups.*

Proof. Let $\delta_1, \delta_2 \in \Gamma_0$ correspond to θ_1, θ_2 as above and the composition $\delta = \delta_1 \circ \delta_2$ in Γ_0 correspond to θ. We have to show that $\theta = \theta_1 + \theta_2$. Recall (More on Morphisms of Spaces, Lemma 04D0) that $\theta_1, \theta_2, \theta$ correspond to derivations $D_1, D_2, D : e_0^{-1}\mathcal{O}_{R_0} \to \mathcal{C}_{U_0/U}$ given by $D_1 = \theta_1 \circ \mathrm{d}_{R_0/U_0}$ and so on. It suffices to check that $D = D_1 + D_2$.

We may check equality on stalks. Let $\overline{u}$ be a geometric point of U and let us use the local rings A, B, C introduced in Section 03DK. The morphisms δ_i correspond to ring maps $\delta_i : B \to A$. Let $K \subset A$ be the ideal of square zero such that $A/K = \mathcal{O}_{U_0,\overline{u}}$. In other words, K is the stalk of $\mathcal{C}_{U_0/U}$ at $\overline{u}$. The fact that $\delta_i \in \Gamma_0$ means exactly that $\delta_i(I) \subset K$. The derivation D_i is just the map $\delta_i - e : B \to A$. Since $B = s(A) \oplus I$ we see that D_i is determined by its restriction to I and that this is just given by $\delta_i|_I$. Moreover D_i and hence δ_i annihilates I^2 because $I = \mathrm{Ker}(I)$.

To finish the proof we observe that δ corresponds to the composition

$$B \to C = (B \otimes_{s,A,t} B)^h_{\mathfrak{m}_B \otimes B + B \otimes \mathfrak{m}_B} \to A$$

where the first arrow is c and the second arrow is determined by the rule $b_1 \otimes b_2 \mapsto \delta_2(t(\delta_1(b_1)))\delta_2(b_2)$ as follows from (08NB). By Lemma 0CK4 we see that an element ζ of I maps to $\zeta \otimes 1 + 1 \otimes \zeta$ plus higher order terms. Hence we conclude that

$$D(\zeta) = (\delta_2 \circ t)\,(D_1(\zeta)) + D_2(\zeta)$$

However, by Lemma 0CKD the action of $\delta_2 \circ t$ on $K = \mathcal{C}_{U_0/U,\overline{u}}$ is the identity and we win. $\square$